WEEK 7

Power BI Report

Student Performance Analysis

1. Introduction

This report presents the Week 7 Power BI analysis of student academic performance. The dashboard summarises student results in three subjects (mathematics, reading and writing) and combines them into an overall average score and an overall pass percentage.

The purpose of the report is to explain what the dashboard shows, interpret the key measures, and draw conclusions that can support teaching and learning decisions.

1.1 Objectives

1. Describe the dataset used in the Power BI report.
2. Present the overall average score and the average score in each subject.
3. Report the overall pass percentage.
4. Compare performance across math, reading and writing.
5. Identify strengths, weaknesses and areas for improvement.

1.2 Tools Used

The analysis was built in Microsoft Power BI Desktop. The report page uses card visuals for the headline measures and a clustered column chart for score comparison. The measures are stored in a table named "Measures table" and calculated over the "StudentsPerformance (2)" table.

2. Dataset Overview

The source table is called StudentsPerformance (2). It contains one row per student, with three numeric score columns.

Column	Description
math score	Student's mark in mathematics
reading score	Student's mark in reading
writing score	Student's mark in writing

2.1 Dataset Summary

Item	Value
Total number of students	[ ]
Lowest score (any subject)	[ ]
Highest score (any subject)	[ ]
Other columns in the table	[ ]

2.2 Data Preparation

Describe any cleaning or transformation done in Power Query, such as removing blanks, correcting data types or renaming columns. If none was needed, state that the data was loaded as provided.

3. Dashboard Overview

The dashboard consists of one page with six visuals: five cards and one chart.

Visual	Type	What it shows
Average	Card	Overall average score
Math Average	Card	Mean math score
Reading Average	Card	Mean reading score
Writing Average	Card	Mean writing score
Overall Pass	Card	Overall pass percentage
Score chart	Clustered column	Overall average score by writing, reading and math score

3.1 Measures

Measure	Purpose
Overall average score	Average of the three subject scores across all students
Average math score	Mean of the math score column
Average Reading score	Mean of the reading score column
Average writing score	Mean of the writing score column
Overall Pass %	Share of students who meet the pass criterion

Pass mark used: [ ] (check the DAX formula of the Overall Pass % measure).

4. Overall Average Score

The overall average score is the headline indicator of student achievement. It combines the three subject results into one figure.

Overall average score: [___]

Interpretation

- An overall average of [___] suggests that students are performing at a [strong / moderate / weak] level.
- Compared with the pass mark of [___], the average is [above / below / close to] the threshold.
- The overall average is [___] points [higher / lower] than the weakest subject.

5. Mathematics Performance

Average math score: [___]

Observations

- The math average is [___] points [above / below] the overall average.
- The lowest and highest math scores are [___] and [___].
- Math is the [strongest / weakest / middle] subject.
- Possible reasons: [difficulty of content, teaching approach, assessment style].

Suggested Actions

- Provide extra practice and revision sessions for students below the pass mark.
- Review topics where scores are consistently low.
- Use peer learning groups to support weaker students.

6. Reading Performance

Average reading score: [___]

Observations

- The reading average is [___] points [above / below] the overall average.
- The lowest and highest reading scores are [___] and [___].
- Reading is the [strongest / weakest / middle] subject.
- Possible reasons: [difficulty of content, teaching approach, assessment style].

Suggested Actions

- Encourage regular reading through classroom libraries and reading logs.
- Use comprehension exercises with short passages.
- Pair reading activities with writing tasks to reinforce both skills.

7. Writing Performance

Average writing score: [___]

Observations

- The writing average is [___] points [above / below] the overall average.
- The lowest and highest writing scores are [___] and [___].
- Writing is the [strongest / weakest / middle] subject.
- Possible reasons: [difficulty of content, teaching approach, assessment style].

Suggested Actions

- Provide structured writing practice with regular feedback.
- Teach grammar and paragraph organisation explicitly.
- Use rubrics so students understand how writing is assessed.

8. Subject Comparison and Pass Percentage

The clustered column chart compares scores across writing, reading and math.

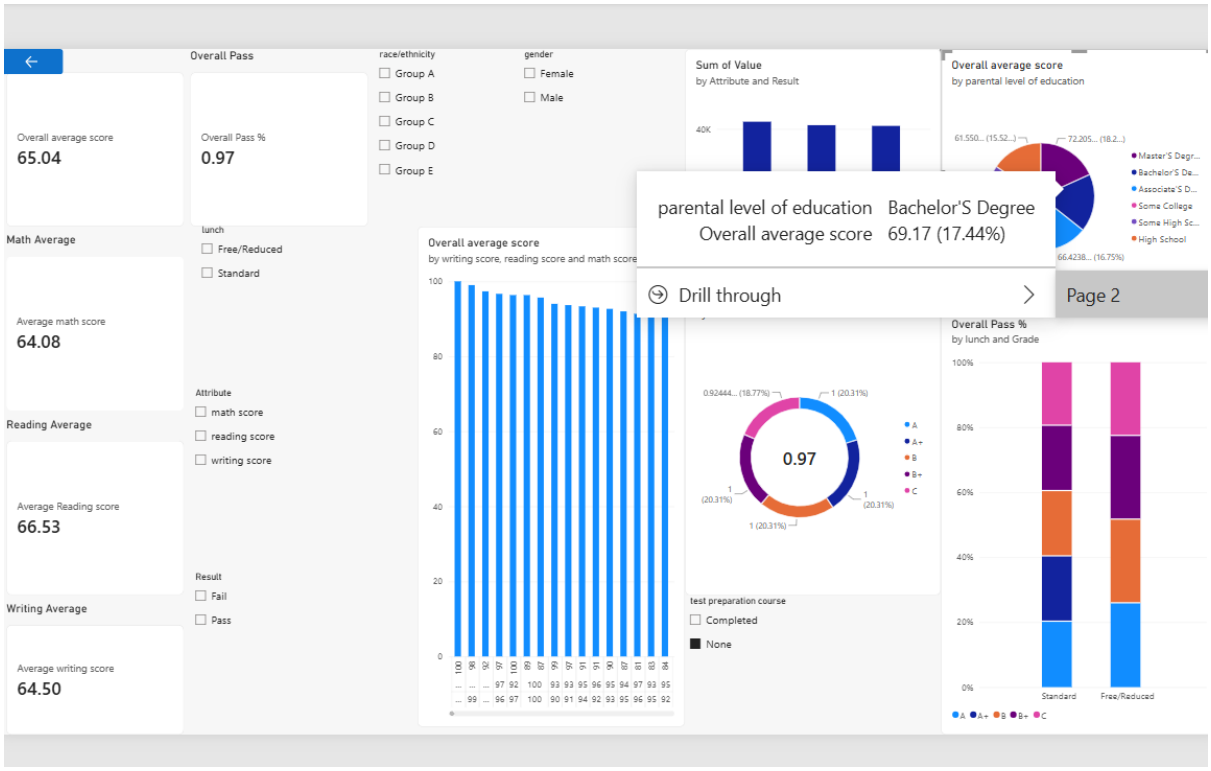
Subject	Average score	Difference from overall
Math	[___]	[___]
Reading	[___]	[___]
Writing	[___]	[___]

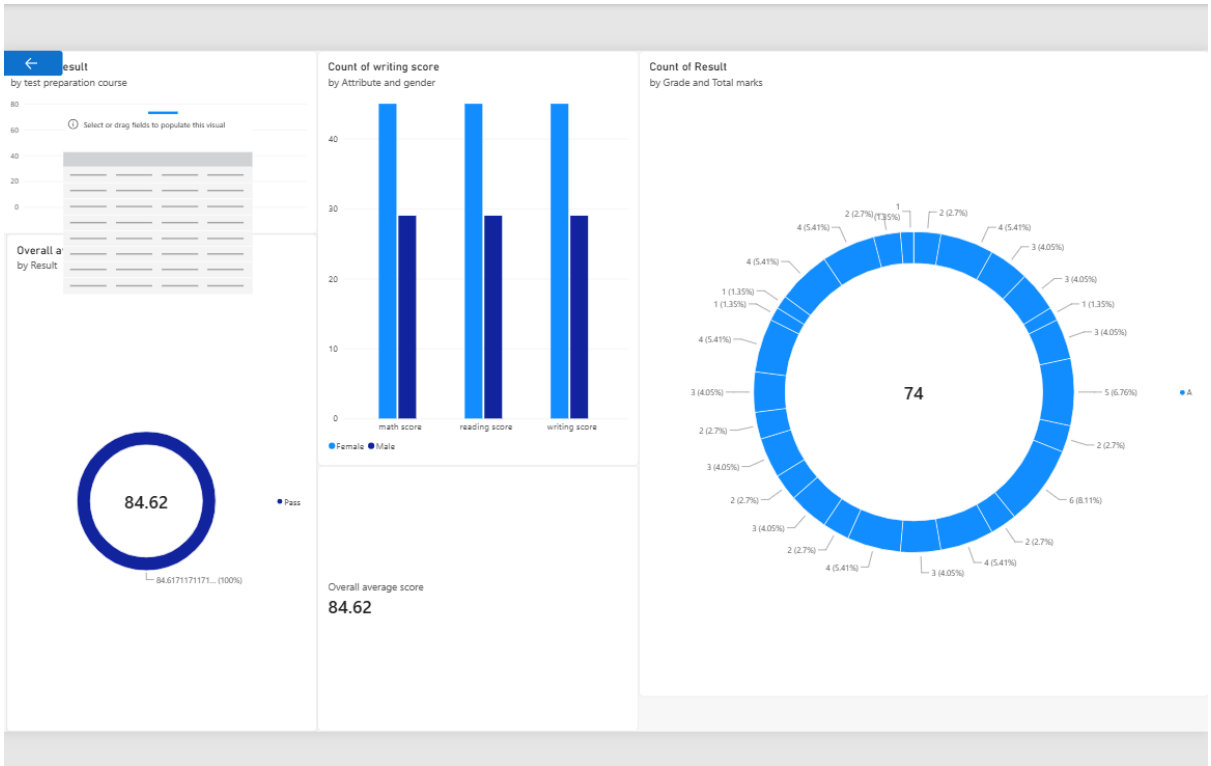
- The strongest subject is [___] and the weakest is [___].
- The gap between the best and weakest subject is [___] points.

Overall Pass Percentage

Overall Pass %: [___]

- [___]% of students passed, so about [___] students did not meet the standard.
- This result is [satisfactory / needs improvement] compared with the target of [___].
- State whether a student must pass every subject or only reach a combined average.


Overall Pass
Overall average score
65.04
Overall Pass %
0.97
race/ethnicity
Group A
Group B
Group C
Group D
Group E
gender
Female
Male
Sum of Value
by Attribute and Result
Overall average score
by parental level of education
parental level of education Bachelor'S Degree
Overall average score 69.17 (17.44%)
Drill through
Page 2
Math Average
Average math score
64.08
lunch
Free/Reduced
Standard
Overall average score
by writing score, reading score and math score
Reading Average
Average Reading score
66.53
Attribute
math score
reading score
writing score
0.97
test preparation course
Completed
None
Overall Pass %
by lunch and Grade
Writing Average
Average writing score
64.50
Result
Fail
Pass
Standard
Free/Reduced


by test preparation course
Count of writing score
by Attribute and gender
Count of Result
by Grade and Total marks
math score
reading score
writing score
Female
Male
84.62
Pass
Overall average score
84.62
74

9. Key Insights and Conclusion

Key Insights

1. The overall average score is [___] and the pass rate is [___]%.
2. [Subject] is the strongest area and [subject] needs the most attention.
3. [Any further pattern, for example a large gap between subjects.]

Conclusion

The Week 7 Power BI dashboard gives a clear, compact view of student performance. Overall results indicate [summary statement]. The subject comparison shows where support should be concentrated, and the pass percentage shows how many students are meeting the standard.